

Chapter 6

Measuring National Output and National Income

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- 2 Calculating GDP
- 3 Nominal versus Real GDP
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Importance of Measurements

- Accurate measures of aggregate variables are the starting point for understanding the economy.
- National Income and Product Accounts (NIPAs)
 - In the 1930s, the Department of Commerce commissioned Simon Kuznets to develop a set of national economic accounts.
 - Data collected and compiled by the [Bureau of Economic Analysis \(BEA\)](#) of the U.S. Department of Commerce
 - to represent and organize the economy and the *sizes* of the various pieces of the economy
 - GDP, GDP Price Deflator, International Transactions (Balance of Payments),

Genesis of GDP

One of the Great Inventions of the 20th Century

- Read the *Economics in Practice* section, p. 115 of the textbook.

Prior to the development of the NIPAs, policy makers had to guide the economy using limited and fragmentary information about the state of the economy. The Great Depression underlined the problems of incomplete data and led to the development of the national accounts:

One reads with dismay of Presidents Hoover and then Roosevelt designing policies to combat the Great Depression of the 1930s on the basis of such sketchy data as stock price indices, freight car loadings, and incomplete indices of industrial production. **The fact was that comprehensive measure of national income and output did not exist at the time.** Depression, and with it the growing role of government in the economy, emphasized the need for such measures and led to the development of a comprehensive set of national income accounts.

Richard T. Froyen

Definition of Gross Domestic Product (GDP)

- GDP: The total market value of all final goods and services produced within a given period by factors of production located within a country.

Note: Key Points of GDP

- ① Final goods and services
- ② Exclusion of used goods and paper transactions
- ③ Exclusion of output produced abroad by domestically owned factors of production

Three Aspects of National Income Identity

- The national income identity says that total income must be equal to the sum of the various components of GDP.

$$\underbrace{\text{GDP}}_{\text{product}} = \underbrace{C + I + G + (EX - IM)}_{\text{expenditure}} = \underbrace{Y}_{\text{income}} .$$

- In equilibrium, what is produced must be consumed, and the distributed income generated through transactions is summed up to the total income in the economy.

The Product Approach

The Valued-Added Approach

- Add the value of all goods and services produced in the economy and then subtract the value of all intermediate goods used in production to obtain total value added.

Exercise: End-of-chapters Problems 1.4

Assume an economy with a lumber producer, a bird house producer, and a craft store. Anissa makes custom bird houses in her garage and she buys all her supplies from a local lumber yard. Last year she purchased \$3,500 worth of supplies and produced 250 bird houses. She sold all 250 bird houses to a local craft store for \$25 each. The craft store sold all the bird houses to customers for \$55 each. For the total bird house production, calculate the value added of a lumber producer, of Anissa, and of the craft store. Calculate GDP in this economy using the product approach.

The Expenditure Approach

The Components of GDP

- The expenditure approach looks at GDP in terms of who *purchases* the produced goods and services.
- Recall the four economic agents in the economy: households, firms, the government, and the rest of the world.
- The expenditure approach calculates GDP by adding these four components of spending.

$$\text{GDP} = \underbrace{C}_{\text{household}} + \underbrace{I}_{\text{firms}} + \underbrace{G}_{\text{government}} + \underbrace{NX}_{\text{foreign}}.$$

The Income Approach

- Note that every payment (expenditure) by a buyer is, at the same time, a receipt (income) for the seller.
- The income approach pays attention to who *receives* it as income.

TABLE 6.3 National Income, 2017

	Billions of Dollars (\$)	Percentage of National Income (%)
National income	16,607.7	100.0
Compensation of employees	10,307.2	62.1
Proprietors' income	1,386.0	8.3
Rental income	743.9	4.5
Corporate profits	2,164.6	13.0
Net interest	586.4	3.5
Indirect taxes minus subsidies	1,268.8	7.6
Net business transfer payments	161.8	1.0
Surplus of government enterprises	-11.0	-0.1

Source: U.S. Bureau of Economic Analysis, March 28, 2018.

Exercise

End-of-Chapters Problems 6.1

Suppose that all income is either compensation of employees or profits. Suppose also that there are no indirect taxes. Calculate GDP from the following set of numbers. Show that the expenditure approach and the income approach add up to the same figure.

Consumption	\$9,500
Investment	3,000
Depreciation	1,750
Profits	2,400
Exports	850
Compensation of employees	11,500
Government purchases	3,200
Direct taxes	1,200
Saving	1,600
Imports	900

Aggregate Output over Time

- We have thus far looked at GDP measured in current dollars, which is nominal GDP—all components of GDP valued at their current prices.
- We come to the following question: if prices of goods and services change year by year, is comparing nominal GDPs a good way to compare the overall living standard each year?

	Quantity	Price per unit
Year 1	100	\$1.00
Year 2	100	\$1.10

- Nominal GDP has increased by \$10 even though no more output was produced.
- The growth in nominal GDP merely stems from the price effect in the above case.

Real GDP

- Real GDP: The production of goods and services valued at constant/fixed (base-year) prices.
- While nominal GDP reflects both the quantities and the prices of the goods and services in the economy, real GDP reflects only the quantities produced by fixing prices constant at base-year levels.
- We can compute GDP deflator, which reflects the overall price level changes.

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100.$$

Exercise

Principles of Macroeconomics, N. Gregory Mankiw

Below are some data from the land of milk and honey.

Year	P_m	Q_m	P_h	Q_h
2020	\$1	100 quarts	\$2	50 quarts
2021	1	200	2	100
2022	2	200	4	100

- 1 Compute nominal GDP, real GDP, and the GDP deflator for each year, using 2020 as the base year.
- 2 Compute the percentage change in nominal GDP, real GDP, and the GDP deflator in 2021 and 2022 from the preceding year.

GDP and Social Welfare

- Does GDP reflect the true meaning of "*well-being*"?
- Some increases in social welfare are associated with a decrease in GDP, or vice versa.
- Complement: green accounting
 - Pollution-cost-adjusted NIPAs
- GDP has nothing to say about the distribution of output among individuals in a society.
- Informal economy
 - Many transactions are missed in the calculation of GDP.